

# Stochastic Simulation Assignment 1

## Assignment C: Limit Order Book

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# 1 Problem and Model Overview

In this Assignment a simplified limit order book was simulated with bid and ask queues and matching engine.

Deterministic formulas that were used for event scheduling:

$$\begin{aligned}\Delta t_n &= 5 (1 + |\sin(n\pi/3)|) && \text{(time between arrivals, s)} \\ p_n &= \text{round}(100 + 2 \sin(ne), 2) && \text{(limit price)} \\ \tau_n &= 30 (1 + \cos^2 n) && \text{(time before cancellation, s)}\end{aligned}$$

Each arrival can be either a limit order with probability 0.7 or a market order with probability 0.3. One run is simulated for 500 trades. Only classification of order to limit (probability 0.7) or market(probability 0.3) and then to bid or ask is random. For all experiments seed 1 was used. Simulation always starts with  $t = 0$  and  $n = 0$ .

## 2 Event Structure

The simulation uses two event types inherited from Des Library: `ArrivalEvent` and `CancelEvent` fired for certain orders.

**Arrival event.** Each arrival can be either a limit order with probability 0.7 or a market order with probability 0.3. Market and limit orders then additionally are classified into bid or ask with probability 0.5 each. Limit order is inserted to book, then its `CancelEvent` is scheduled and then match method is called. Market order is immediately filled with the best opposite order or is rejected if opposite side is empty. The next arrival event is always scheduled at  $t + \Delta t_n$  time.

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**Algorithm 1** Arrival event ( $n$ -th order, time  $t$ )

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1: if  $U_1 < 0.7$  then ▷ limit order
2:   side  $\leftarrow$  bid or ask, each with probability 1/2
3:   insert order with price  $p_n$  into respective queue
4:   schedule CANCELEVENT at  $t + \tau_n$ 
5:   MATCH( $t$ )
6: else ▷ market order
7:   side  $\leftarrow$  bid or ask, each with probability 1/2
8:    $o \leftarrow$  best resting order on the opposite side
9:   if  $o$  exists then
10:    FILL( $o$ ); trades  $\leftarrow$  trades +1
11:   else
12:    rejections  $\leftarrow$  rejections +1
13:   end if
14: end if
15: if trades  $\geq 500$  then
16:   stop simulation
17: else
18:   schedule next ARRIVALEVENT at  $t + \Delta t_n$ 
19: end if
```

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**Cancellation event.** Every order can leave the book only by match or by cancellation. Each limit order gets its own `CancelEvent` that fires on time or is removed if order is filled. `CancelEvent` only executes for an order that still is in the book and matching respective order removes it's `CancelEvent` by using des library `event.cancel()` function.

**Matching engine.** After every arrival and cancellation event a match function is called which checks current best bid and best ask and, if best bid  $\geq$  best ask, removes both limit orders. It is enough to match only after these events because the book is uncrossed before and one insertion creates at most one cross which is removed by match if it exists.

### 3 Data Collection

Des library statistics objects were used to gather statistics. Each one was updated only when event happened that can change its value.

- **Time-averaged spread:** TimeWeightedStatistic object that accumulates  $s_t = a_t - b_t$  when both  $a_t, b_t$  are defined. Additional TimeWeightedStatistic object is used to gather whether both  $a_t, b_t$  were defined to later get a true spread by dividing first one by second one.
- **Average time-to-fill:** SampleStatistic object accumulating  $t_{\text{match}} - t_{\text{arrival}}$  for each limit order that finds a match.
- **Cancellation fraction:** number of canceled limit orders divided by the total number of limit orders.
- **Average queue lengths:** Two TimeWeightedStatistic objects that accumulate each queue length.

### 4 Results and Interpretation

The simulation runs up to  $T = 12,033$  s, which is the time of the 500th trade.

| Metric                       | Value |
|------------------------------|-------|
| Time-averaged bid ask spread | 1.87  |
| Average time-to-fill (s)     | 12.9  |
| Cancellation fraction        | 0.295 |
| Average bid queue length     | 0.875 |
| Average ask queue length     | 1.036 |
| Rejected market orders       | 214   |

Table 1: Metrics over  $[0, T]$ .

Spread is expected to be positive by construction (matching removes every crossing that appears). Average queue length differences can be explained by relatively small time frame. With more trades, difference becomes smaller which was checked with longer runs. Additionally both queues average length is near 0.95 which explains why there are so many rejections and high cancellation fraction.